Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/01/2019
PI Name:	Hogan, Harry Ph.D.		
Project Title:	Pre-Treatment Approaches for Improving the Response of Bone to Disuse and Recovery		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	hhogan@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-845-1538
Organization Name:	Texas A&M University		
PI Address 1:	Dept. of Mechanical Engineering		
PI Address 2:	TAMU 3123		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-3123	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2018 per NSSC information (Ed., 8/22/17) NOTE: End date changed to 8/31/2017 per NSSC information (Ed., 6/3/16) NOTE: Period of performance dates corrected per NSSC information (Ed., 9/10/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University) Bloomfield, Susan Ph.D. (Texas A&M University) Fluckey, James Ph.D. (Texas A&M University)		
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Task Description:	This research proposal is aimed at better understanding the basic bone biological effects of various osteoporosis treatments as therapies for reducing or preventing bone loss due to space flight. The project will study hindlimb unloaded (HU) rats, which is a technique that is well established as an effective and informative ground-based analog for studying the effects of space flight on the musculoskeletal system. In order to better understand the detailed effects of each, the treatments will be studied separately. The four treatments to be examined include two anabolic (bone building) therapies and two anti-resorptive therapies. One anabolic approach will use voluntary exercise and the other a new osteoporosis drug (anti-sclerostin antibody). Anti-resorptive drugs are the most popular for treating clinical osteoporosis and achieve their benefits by preventing bone resorption, which is part of the normal cycle of bone replenishment. Furthermore, the treatments will be used as pre-treatments and only administered prior to HU exposure. This is not realistic for astronauts, but it will allow more in-depth study of the fundamental nature of each treatment, and it will also align with the anticipated requirements of possible follow-up space flight experiments using the same rat animal model. The experiments to be conducted will characterize beneficial effects of pre-treatment upon bone loss due to HU as well as bone recovery following HU and return to weightbearing. Bone properties will be quantified using CT-based analyses, histological techniques for assessing bone cell function, as well as biomechanical testing to measure strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of the studies outlined in this research proposal will provide new insights into the basic nature of the effects of exercise and pharmaceutical agents and how these interact with mechanical unloading and recovery therefrom. Removal of weightbearing through hindlimb unloading is a severe and acute form of mechanical disuse that has been shown to simulate many of the effects of microgravity exposure due to space flight on the musculoskeletal system. Hindlimb unloading also induces many musculoskeletal system changes that are similar to accelerated aging as well as other disuse scenarios such as stroke, spinal cord injury, or neuromuscular impairment. Thus, insights gained should also help advance basic understanding of the fundamental biology associated with Earth-bound clinical medicine as well as space flight effects. Future space flight experiments can be designed to replicate the research to be conducted in this proposed project, and comparing results from the two will provide essential and more detailed understanding of the full efficacy, as well as limitations, of the adult rat HU model as a ground-based analog for human space flight. Studying these treatments separately is not necessarily realistic for actual mission scenarios, but the results will nevertheless provide evidence-based rational data for contemplating and designing possible co-therapies using either combinations (simultaneously) or possible sequential scheduling protocols.
	As explained in previous year progress reports, an adjustment was made during year 1 to reverse the order of the two animal experiments. In the proposal, the first set of animal studies (Experiment 1) was to address Specific Aim 1, which compares two anabolic pre-treatments, but we decided to change this to address Specific Aim 2 first. A major consideration driving this change was to allow more time to acquire the Sclerostin-Antibody drug. Four experimental groups were needed for each of the two sets of animal experiments as further described below. Anti-Resorptive Pre-Treatment Animal Work: The two anti-resorptive pre-treatments are both bisphosphonate drugs, specifically risedronate and alendronate. The overall protocol structure can be defined as follows: Pre-treatment (PT) period = days 0 to 28 Hindlimb unloading (HU) period = days 28 to 56 Ambulating recovery (AR) period = days 56 to 112 The four groups are defined as follows: Control (CON) = normal cage activity for PT, HU, and AR periods HU Control (HUC) = normal cage for PT, unloading during HU, normal activity for AR Risedronate (RIS) = RIS for PT, unloading during HU, normal activity for AR Alendronate (ALN) = ALN for PT, unloading during HU, normal activity for AR The animal experiments were conducted in three cohorts of animals due to constraints and considerations associated with animal housing space, animal cages, and personnel requirements. Each cohort involved 45-50 animals, with the scheduling timetable as follows: Cohort 1 – 27 May 2014 to 9 Oct 2014 Cohort 2 – 15 July 2014 to 25 Nov 2014 Cohort 3 – 30 Sept 2014 to 10 Feb 2015 All animal experiments included in vivo measures that were made on all animals every 28 days. The main assessment conducted was pQCT (peripheral Quantitative Computed Tomography) scans, which are made at the proximal tibia metaphysis (PTM) and tibia diaphysis (TD) of anesthetized animals. In addition, blood was drawn at 28-day intervals for possible serum analysis of bone activity biomarkers. Anabolic Pre-Treatment Animal Work: The four groups identified for the anabolic pre-treatment studies are defined below along with a schematic depiction of the animal protocols. The two anabolic pre-treatments are pharmaceutical administration and mechanical loading through exercise. The specific pharmaceutical agent used was an anti-sclerostin anti-body compound provided by Lilly (Eli Lilly and Company, Indianapolis, IN) through a Material Transfer Agreement. The mechanical loading anabolic treatment consisted of a voluntary rat jumping protocol intended to simulate resistance exercise. The overall protocol structure is again defined as follows: Pre-treatment (PT) period = days 0 to 28 Hindlimb unloading (HU) period = days 28 to 56 Ambulating recovery (AR) period = days 56 to 112

Task Progress:

The four groups are defined as follows:

Control (CON) = normal cage activity for PT, HU, and AR periods

HU Control (HUC) = normal cage for PT, unloading during HU, normal activity for AR

Sclerostin Anti-body (SCL) = SCL for PT, unloading during HU, normal activity for AR

Exercise (REX) = REX for PT, unloading during HU, normal activity for AR

The same in vivo measures were made on all animals every 28 days; specifically, pQCT (peripheral Quantitative Computed Tomography) scans were made at the PTM and TD of anesthetized animals, and blood was drawn at 28-day intervals for possible serum analysis of bone activity biomarkers.

Major milestones for the anabolic pre-treatment portions of the project are summarized as follows:

July – December 2015: • Material Transfer Agreement (MTA) completed July 2015 ; • Receipt of initial shipment of Scl-Ab October 2015

November – December 2015: • Pilot study for Scl-Ab dosing ; • Pilot study for voluntary jumping exercise

January – September 2016:

• Animal work for Scl-Ab drug pre-treatment conducted in 3 cohorts:

Cohort 1 – 26 Jan 2016 to 22 June 2016 (20 Scl-Ab; 5 REX)

Cohort 2 – 18 Apr 2016 to 31 Aug 2016 (13 Scl-Ab)

Cohort 3 – 5 May 2016 to 14 Sept 2016 (12 Scl-Ab)

September – December 2016: • Continued refinement of voluntary jumping exercise details and new cages

January 2017 – April 2018: • Animal work for voluntary jumping exercise (REX) conducted in 4 cohorts Cohort 1 – 10 Jan 2017 to 16 May 2017 (10 REX)

Cohort 2 – 13 June 2017 to 14 Nov 2017 (10 REX)

Cohort 3 – 13 Sept 2017 to 10 Feb 2018 (10 REX)

Cohort 4 – 12 Dec 2017 to 8 Apr 2018 (10 REX)

As reflected above, a decision was made to stagger the two parts of the anabolic pre-treatment arm of the project. The Scl-Ab (pharmacologic) pre-treatment part was done first and was completed in September 2016. The voluntary jumping exercise (REX) pre-treatment part was resumed in January 2017 and continued into to 2018. Two factors prompted the strategy to stagger the Scl-Ab and REX parts of the study. First, the duration of the window of availability for the sclerostin anti-body compound was uncertain, so it was deemed best to acquire all of the material that was promised as soon as possible and reasonable. The second factor was that the voluntary jumping exercise required additional refinements and adjustments in order to make it practical and effective for the remaining 40 animals scheduled to conduct the jumping exercises. Five animals were included in the first cohort of the Scl-Ab pre-treatment part, and the knowledge gained from that was used to adjust and improve the process. Most prominently, the customized cages required for acclimating, training, and exercising the animals needed to be redesigned and new ones constructed. Consequently, the first of the four remaining REX cohorts was not started until January 2017. In addition, the labor-intensive nature of the jumping exercise training protocol dictates a maximum of 10 animals per cohort, as reflected in the summary above.

An overall summary of the major in vivo and ex vivo measurements is as follows.

In Vivo Longitudinal Measures

Densitometry by pQCT (left tibia; proximal metaphysis and midshaft) ; ~ vBMD, BMC, cross-sectional area, cross-sectional moment of inertia, etc.

Ex Vivo Endpoint Measures

Densitometry by pQCT (tibia; proximal metaphysis and midshaft): ~ vBMD, BMC, cross-sectional area, cross-sectional moment of inertia, etc.

Densitometry and microstructure by microCT (tibia and femur; metaphyses, midshaft, FN) ; ~ trabecular microarchitecture, vBMD, BMC, TMD, etc.

Static and dynamic histomorphometry (tibia and femur; metaphyses, midshaft, FN): ~ trabecular, cortical, MS/BS, MAR, BFR, etc.

Bone Strength (tibia metaphysis RPC, tibia & femur midshaft, FN): ~ ultimate load, stiffness, ultimate strength, elastic modulus, toughness, etc.

Muscle Weights (hindlimb posterior crural muscles): ~ wet weights for gastrocnemius, soleus, and plantaris

All in vivo pQCT measurements have been made and data analyzed.

Almost all of the ex vivo measurements have been completed as well.

Results have been disseminated to date primarily through conference abstracts, posters, and presentations, as indicated in the Bibliography section. One journal article has been published.

Two PhD students are finishing their dissertations in fall 2019 (Jessica Brezicha) and spring 2020 (Scott Lenfest). These dissertation research projects focus on the two anabolic pre-treatments, sclerostin anti-body and voluntary jumping exercise, respectively. Because of the sequential and inter-dependent nature of the animal experiments, outcome measures gathered, and subsequent data analyses, dissemination through journal articles has been delayed. Publications

	should be readily forthcoming from the completed dissertation work of these two PhD students. Publication of the earlier results for the two bisphosphonate drug pre-treatments is also being coordinated with publication of the anabolic pre-treatment results to permit detailed comparisons and more in-depth consideration.
Bibliography Type:	Description: (Last Updated: 01/11/2021)
Abstracts for Journals and Proceedings	Elizondo JP, Black JB, Brezicha JE, Lenfest SE, Kosniewski JL, Bloomfield SA, Allen MR, Hogan HA. "Comparison of Bisphosphonate Pre-Treatments on Simulated Microgravity Unloading Effects at the Tibia and Femur in Adult Rats." 44th European Calcified Tissue Society Congress, Salzburg, Austria, May l3-16, 2017. Poster Presentation. Musculoskeletal. 44th European Calcified Tissue Society Congress, Salzburg, Austria, May l3-16, 2017. , May-2017
Abstracts for Journals and Proceedings	Lenfest SE, Koskniewski JL, Metzger CE, Brezicha JE, Elizondo JP, Looper AL, Bloomfield SA, Hogan HA. "Voluntary Jumping Exercise in Rats Prior to Unloading Prevents Unloading-Related Bone Loss." ASBMR 2017 (American Society for Bone and Mineral Research) Annual Meeting, Denver, CO, September 8-11, 2017. J Bone Miner Res 2017;32 (Suppl 1):SA0363. , Sep-2017
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Abstracts for Journals and Proceedings	Elizondo JP, Lenfest SE, Metzger CE, Brezicha JE, Kosniewski JL, Looper, AL, Bloomfield SA, Hogan HA. "Characterizing Voluntary Jumping Exercise in Rats as a Countermeasure to Subsequent Unloading." Poster Session A: Bone. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Poster #19185. , Jan-2019
Abstracts for Journals and Proceedings	Elizondo JP, Brezicha JE, Lenfest SE, Sihra S, Kosniewski JL, Black JB, Kohn Z, Bloomfield SA, Hogan HA. "Pretreatment with Anti-Sclerostin Antibody has Lasting Osteogenic Effects on the Femur of Unloaded Male Rats." Plenary Poster, ASBMR 2019 (American Society for Bone and Mineral Research) Annual Meeting, Orlando, FL, September 20-23, 2019. J Bone Miner Res 2019;34(Suppl 1). Poster Presentations FRI-0026 & SAT-0026. , Sep-2019
Articles in Peer-reviewed Journals	Metzger CE, Brezicha JE, Elizondo JP, Anand Narayanan S, Hogan HA, Bloomfield SA. "Differential responses of mechanosensitive osteocyte proteins in fore- and hindlimbs of hindlimb unloaded rats." Bone. 2017 Dec;105:26-34. Epub 2017 Aug 3. https://doi.org/10.1016/j.bone.2017.08.002 ; PubMed PMID: 28782619 , Dec-2017
Dissertations and Theses	Black J. (Jeremy Black) "Comparison of the Persisting Effects of Bisphosphonate Treatments Prior to Hindlimb Unloading on Mechanical and Densitometric Properties in the Tibia of Adult Male Rats." MS Thesis, Texas A&M University, December 2016. , Dec-2016
Dissertations and Theses	Kosniewski J. (Jennifer Kosniewski) "The Efficacy of Bisphosphonate Pre-treatment in Preventing Losses in Densitometric and Mechanical Properties During Hindlimb Unloading and Throughout Reambulation in the Distal Femur Metaphysis of Adult Male Rats." MS Thesis, Texas A&M University, May 2017. , May-2017
Dissertations and Theses	Brezicha J. (Jessica Brezicha) "Skeletal Responses to Simulated Microgravity in the Adult Rat after Anti-Sclerostin Antibody Pretreatment." PhD Dissertation, Texas A&M University, December 2019. , Dec-2019
Dissertations and Theses	Lenfest S. (Scott Lenfest) "Skeletal Effects of Bisphosphonate Treatment and Simulated Resistance Exercise on Adult Rats Exposed to Simulated Microgravity. " PhD Dissertation, Texas A&M University, May 2020. , May-2020